

SOHAM SONAR

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EDUCATION

Master of Computer Science

August 2023 - May 2025

Illinois Institute of Technology, Chicago, IL

Relevant Coursework: Machine learning, Cloud computing, Design and analysis of algorithms, Advanced operating systems

Bachelor of Computer Engineering

August 2018 - July 2022

Savitribai Phule Pune University

Relevant Coursework: Data structures and algorithm, Object oriented programming, Advanced database organization, Big data

EXPERIENCE

Founding Software Engineer

December 2025 - Present

Sandman CI

Chicago, IL

- Spearheaded end-to-end development owning product definition, UX, frontend, backend, cloud infrastructure, and deployment to ship an **AI analytics platform**, architected for multi-tenant scale (500+ users).
- Designed and implemented a production-grade LLM system combining structured analytics, **retrieval-augmented generation**, and prompt versioning to analyze **over 150+ datapoints**, while improving response grounding and consistency.
- Built scalable platform foundations (Next.js, Supabase/Postgres, Vercel, Stripe, CI/CD) with **secure auth, subscription billing, streaming chat, and cost controls**, enabling controlled growth from early pilots (15 users) to 150+ users.

Research Assistant

March 2025 - September 2025

Gnosis Research Center - Illinois Institute of Technology

Chicago, IL

- Developed an integrated **Agentic AI platform** leveraging multi agent orchestration to automate workflows across 40+ node clusters, enabling **autonomous task execution** and **intelligent operations coordination**.
- Enhanced the performance of **open source projects (IOWarp, Chronolog)**, by designing **REST APIs** and integrating an intuitive **assistant** for **data analytics** and **AI driven workflows**, reducing average data retrieval latency by 60%.
- Automated **CI/CD pipelines** with **GitHub Actions** and **Docker**, automating build, lint testing, and deployment processes across on prem systems and scalable cloud environments for faster and more reliable application delivery.

Executive

March 2023 - June 2023

Hexaware Technologies

Mumbai, India

- Architected and debugged software applications for healthcare solutions using **Python**, and **SQL**, achieving 60% performance improvement through **software architecture optimization**.
- **Streamlined data entry workflows**, reducing manual workload and improving data processing efficiency by 30% through scripting and **workflow automation**.
- Collaborated with **cross functional** teams in an **Agile Scrum** environment, and led backlog grooming and sprint planning across software engineering and QA teams, reducing post-deployment defects by 30%.

SKILLS

Programming Languages: Python, Typescript, Java, C++, SQL, HTML, CSS, Bash/Shell Scripting.

AI Development: Autogens, FAISS, Finetuning, Github Copilot, Langchain, Vector DB, LLMs, RAG Models, Claude.

Machine Learning: Tensorflow, Pytorch, Scikit-learn, Huggingface, Keras, supervised & unsupervised models, MLflow.

Web/Software Development: Angular, React, Next.js, REST API, FastAPI, JIRA, Linux, Git, Agile, Scrum, VS Code.

Cloud/Big Data & Database: AWS, GCP, Azure, Docker, Kubernetes, Hadoop, Spark, Kafka, MySQL, PostgreSQL.

PERSONAL PROJECTS

Docquery [\[Link\]](#)

- Engineered an **AI powered knowledge** and document querying platform using **FastAPI**, **PostgreSQL**, **FAISS**, and **OpenAI embeddings**, enabling hybrid semantic-keyword retrieval with 30% higher answer accuracy than regular systems.
- Implemented scalable, production grade architecture with **Redis caching**, **async processing**, **supporting 200+ concurrent users** while reducing compute cost per query by 40% through intelligent caching and incremental indexing.

Enterprise IO Automation Framework [\[Link\]](#)

- Led the development of the **Scientific Model Context Protocol (MCP) server framework**, including Pandas, Parquet, Plot and HDF5 MCP servers, to **automate I/O** and **filesystem workflows** for local and cloud environments.
- Designed a **custom LLM client** using **Google Gen AI sdk** to coordinate **120+ simulation pipelines**, processing **multi-terabyte datasets** and significantly **reducing data access latency** across distributed systems.